

Generation Interconnection System Impact Study Report

For Queue Project Q22-1996

Metaline Falls 100 kV

February 29, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-1996. The Interconnection Customer proposes a 104.8 MW battery energy storage facility interconnecting at the Metaline Falls 100 kV substation in the Ohio area. The request was received on February 04, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 448 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,357,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WILBRAHAM to YORBA LINDA 100.0 kV
Contingency: PERRY to ARGENTA 100.0 kV
Loading: 136.1% (Rating: 117 MVA)
DFAX: 32.3%
Pre-project loading: 107.2%
2. Facility: ARGENTA to PERRY 100.0 kV
Contingency: WILBRAHAM to YORBA LINDA 100.0 kV
Loading: 113.8% (Rating: 144 MVA)
DFAX: 21.5%
Pre-project loading: 98.2%
3. Facility: FT BRAGG 230/100kV XFMR Bank 1
Contingency: FT BRAGG 230 kV/100 kV TX
Loading: 113.1% (Rating: 161 MVA)
DFAX: 52.2%
Pre-project loading: 79.1%
4. Facility: FOT BRAGG 230/100kV Autotransformer Unit 2
Contingency: FORT BRAGG 230 kV/100 kV Transformer #1
Loading: 113.1% (Rating: 161 MVA)
DFAX: 52.2%
Pre-project loading: 79.1%
5. Facility: WALTERIA to PERRY SUBSTATION 100.0 kV
Contingency: PERRY to WALTERIA SUBSTATION 100.0 kV Cir. 2
Loading: 104.7% (Rating: 146 MVA)
DFAX: 7.6%
Pre-project loading: 99.3%
6. Facility: PERRY to WALTERIA SUBSTATION 100.0 kV #2
Contingency: WALTERIA to PERRY 100.0 kV Cir. 1
Loading: 104.7% (Rating: 146 MVA)

DFAX: 7.6%
Pre-project loading: 99.3%

7. Facility: CASE to TWINSBURG 100.0 kV
Contingency: RAVNNA to RUDD SUBSTATION 100.0 kV
Loading: 103.2% (Rating: 124 MVA)
DFAX: 6.7%
Pre-project loading: 97.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WILBRAHAM to YORBA LINDA 100.0 kV	Rebuild line with higher-capacity conductor	\$3,663,000	100% (\$3,663,000)
ARGENTA to PERRY 100.0 kV	Rebuild line with higher-capacity conductor	\$1,450,000	100% (\$1,450,000)
FT BRAGG 230/100kV XFMR Bank 1	Replace transformer with higher-rated unit	\$6,975,000	100% (\$6,975,000)
FORT BRAGG 230-100 kV Auto Unit 2	Replace transformer with higher-rated unit	\$11,459,000	100% (\$11,459,000)
PERRY to WALTERIA 100.0 kV Line 1	Reconductor line	\$1,067,000	100% (\$1,067,000)
PERRY to WALTERIA SUBSTATION 100.0 kV #2	Reconductor line	\$1,075,000	100% (\$1,075,000)
CASE to TWINSBURG 100.0 kV L	Reconductor line	\$668,000	100% (\$668,000)
	Total allocated to project		\$26,357,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.